

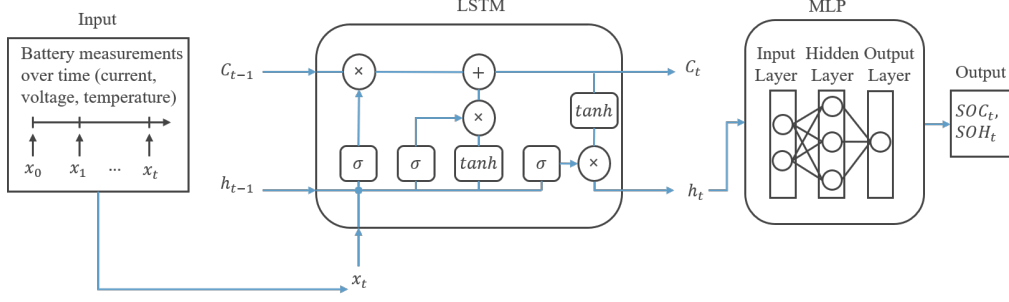


Figure 7.1: Stateful LSTM + MLP estimator structure used for both SOC and SOH estimation. Preprocessed feature streams are processed step by step by a single-layer LSTM whose hidden and cell states are propagated across time. A lightweight MLP head then maps the hidden representation to either SOC or SOH. The SOC instance uses hidden size 64 with an MLP width of 64 and a sigmoid output, the SOH instance hidden size 128 with an MLP width of 128 and a linear output. This structure is the basis for the later pruning, quantization, and MCU export steps.

7.3 LSTM Architecture and Training Setup

Both SOC and SOH estimators employ the LSTM update defined in Equation 3.12, but with task-specific feature sets and sizes, as summarized in Figure 7.2. The SOC model uses a single-layer LSTM with hidden size $H = 64$ that consumes a six-dimensional feature vector $\{U, I, T, Q_c, dU/dt, dI/dt\}$. Its MLP head contains one ReLU-activated hidden layer of width 64 and a sigmoid output neuron that constrains the estimate to $[0, 1]$. The SOH model uses the same overall structure with hidden size $H = 128$ and an MLP hidden width of 128. It consumes the feature vector $\{t, U, I, T, EFC, Q_c\}$ and ends in a linear output neuron because the slowly varying SOH target does not benefit from a saturating output activation.

In contrast to the hourly aggregated SOH estimator of Chapter 6, both models operate directly on the sample-wise 1 Hz stream. The hidden and cell states are propagated continuously between samples. The implementation follows the standard PyTorch gate order and reproduces the same matrix operations, state update, and activation functions in C.[32, 59] The application-specific architecture in Figure 7.1 therefore extends the generic cell in Figure 3.3 by adding the six-channel feature input and the task-specific MLP output heads.